

eDNA Analysis of Biodiversity in Manoa Stream

Introduction

This eDNA analyses was conducted on a sample of stream water collected by a stream biologist who wants to determine if eDNA can provide an alternative to manual stream surveys. Water was collected from Manoa stream near Manoa Marketplace. In this project in particular, we used the primer, 12S rDNA, to search for samples of the Rainbow Guppy (*Poecilia Reticulata*). The Rainbow Guppy is a freshwater fish that lives in small springs and streams across the islands on Hawaii. Introduced in Hawaii through fish farms in 1922, it became an invasive species that eats the eggs of native animals, such as damselfishes on O'ahu (DLNR, 2021). Because of the introduction of invasive species, it also introduces the threat of a declining ecosystem where it becomes susceptible to collapsing. One survey found that 50-90% of the fish community in sampled wetland sites were invasive species from the family, Poeciliidae, which includes guppies (CTAHR, 2009). This shows how invasive species are able to reproduce quickly to outcompete native species food and habitat. It is therefore important to find out the extent of the Rainbow Guppy's impact on the biodiversity of streams in Hawaii. It will give researchers an idea of what must be done to protect the biodiversity of ecosystems in Hawaii.

Sample collection and sequencing preparation

To sample the biodiversity in the Manoa stream, 1L of water was collected and split into 2 replicates of 400mL of and centrifuged at 4500 rpm for 15 minutes to pellet the organic material. DNA was extracted from each replicate using the NEB Monarch kit following the manufacturer's protocol and eluted in 30uL of elution buffer. The 12S region of fish species was amplified through PCR using the MiFish primers developed by Miya et al. (2015). Following PCR amplification, the sequences were prepared for sequencing using the Native Barcoding kit (SQK-NBD-104). Barcoded libraries quantified using a Qubit fluorometer, normalized based on DNA concentration and pooled in equimolar amounts. The pooled library was then sequenced on a single Oxford Nanopore MinION Flongle (FLO-FLG001) for 72 hours. Reads were basecalled and adaptors and barcodes were trimmed using the GUPPY v3.2.2 high accuracy model.

Classification of eDNA reads

First, we filtered the survey data for vertebrates and compiled a list of all unique vertebrate sequences detected in the sample. To identify each sequence, we located the corresponding 12S rDNA accession numbers through GenBank and used these for analyses. All bioinformatic processing was completed using the Decona pipeline (Oosterbroek et al., preprint; <https://github.com/Saskia-Oosterbroek/decona>) within a Docker environment, in combination with NCBI-BLAST 2.9.0+ for taxonomic assignment. Amplicon reads were initially quality-filtered with NanoFilt v2.7.1, retaining only those with a minimum Q-score >10 and lengths between 170–300 bp (De Coster et al., 2018). Sequences appearing more than five times and sharing >95% identity based on a k-mer length of 5 were clustered using CD-HIT (Fu et al., 2018). Clustered sequences were then aligned using Minimap2 v2.17-r941 (Li, 2018). Initial consensus sequences were generated with Racon v1.4.21 (Vaser et al., 2017) and polished using Medaka v1.1.2 (Oxford Nanopore Technologies, 2018). Final consensus sequences were compared against a custom local BLAST database consisting of publicly available fish 12S and full mitochondrial sequences downloaded from GenBank (accessed September 15, 2022). Sequences returning >95% identity were assigned to the corresponding fish species, while sequences <95% identity were designated as unclassified.

```

# pull the docker image
docker pull ethill/decona:latest
# run the docker image
docker run \
  --name= \
  --volume=:/home/data \
# change to the directory with the data

# command for the decona pipeline
decona
# -f = input file
# -l = minimum length of read
# -m = maximum length of read
# -q = minimum quality score
# -c = minimum percent identity
# -n = minimum number of clustered reads to call a consensus
# -k = kmer length
# -T = number of threads
# -B = blast database

```

Making the environment

To begin this analysis we will load the packages required for our manipulations and visualizations. Then a accessory function was created to check if an integer is of 0 length. In addition, we created a function that found the number of species in a sample.

```

# Load the required packages
library(stringr)
library(ggplot2)
library(dplyr)
library(reshape2)
#setwd("edna_project")
# Accessory function:
# Checks if an integer is of 0 length, you need to complete this function!
is.integer0 <- function(x){
  if(length(x) == 0L){
    return(TRUE)
  }else{
    return(FALSE)
  }
}

# Rarefaction function, you need to complete this function!
rare_sample <- function(x, y) {
  x <- x[sample(nrow(x), y, replace = FALSE), ]
  # how to remove duplicates?
  #x <- x[!duplicated(x$gensp), ]

  unique_x <- unique(x$gensp)
  # how do we find the number of unique species?
  num_unique <- length(unique_x)
  # what do we want to return?

```

```

    return(num_unique)
}

```

```

all <- read.csv("./all.csv")

# str(me)

filtered_all <- all[all$OrganismType == "Vertebrate" ,]

filtered_all <- filtered_all[!is.numeric(filtered_all) ,]

#head(filtered_all)

unique_species <- unique(filtered_all$Species)

#unique_species

```

Ingesting the data

We then loaded our eDNA classification as well as the survey data collected from the same date as the eDNA sample. We combined the two barcodes and removed any rows that did not have any data and fixed a typo in the survey data before it caused any problems downstream. The typo, “Xiphophorus helleri” was changed to “Xiphophorus hellerii”.

```

# How do we read in the data considering we have to read in multiple files?
decona_out <- list.files("./result/Racon")
#read.csv("file.txt", sep = "\t")
blast_out <- decona_out[grep("summary_BLAST", decona_out)]
# decona_out

# blast_out

# Prep the data frame
sdat <- data.frame()

#raw text output
for(i in blast_out){
  dat <- read.delim(file.path("result/Racon", i), sep = "\t", header = FALSE, skip = 1, col.names = paste0("V", 1:12))

  check <- grep("TRUE", is.na(dat$V1))

  #if there is not clusters, return an empty data frame
  if(is.integer0(check)== TRUE){
    dat <- dat[-c(grep("TRUE", is.na(dat$V1))),]
    temp <- dat[, c(1,4,8,11,12)]
  }else {
    temp <- dat[, c(1,4,8,11,12)]
  }

  # Read in the classification tables from decona

```

```

# HINT: you will need to pad table with additional columns to ensure all tables have the same number of
# HINT: what programming construct can you use to loop through the files? What object will you be looping
# HINT: below is the general outline of what you will need to do

```

```

# dat <- # Reads in the data, use ?read.delim
#check <- # How can we check if there are any NA values in the first column?

names(temp) <- c("reads", "percent_id", "e_value", "genus", "species")
temp$percent_id[is.na(temp$percent_id)] <- 0

#names(temp) <- paste(temp$genus, temp$species, sep = "_")
temp$barcode <- str_extract(i, "barcode[0-9]+")

sdat <- rbind(sdat, temp)
}

sdat$gensp <- paste(sdat$genus, sdat$species, sep = "_")

```

```

# If there are no clusters, return an empty data frame
# How can we write a test to ID clusters of length 0 and remove the row?
# HINT: you will need to use the is.integer0 function
# HINT: The columns you are going to keep are 1, 4, 8, 11, 12.
# if a cluster is of length zero, we will want to remove the row

# Rename the columns to the following:
# "reads", "percent_id", "e_value", "genus", "species"
# Any clusters with a NA percent_id should be replaced with 0

# Add column for gensp and barcode
#temp$gensp <-
# Add column for barcode
# HINT: use ?str_extract
#temp$barcode <-

# Combine the dataframes
# sdat <- rbind(sdat, temp)

# Read in the survey data
sur <- read.csv("./me.csv", sep = ",", header = TRUE)
# Fix a typo HERE BEFORE I FORGET
sur$Species[sur$Species == "Xiphophorus helleri"] <- "Xiphophorus hellerii"

```

Once the data was cleaned up, we renamed the columns to be more descriptive and removed any rows that were not of interest. Specifically, the rows that were identified as humans, pigs and turtles were removed since we are only looking at a sample of fish in the Manoa Stream. Classifications that were returned with a percent identity of less than 95 were removed because a lower percent identity indicates a less reliable match to a known sequence in the database. Additionally, we also created a new column that combined the genus and species columns to make it easier to work with downstream.

```

# Remove non-fish classifications
# HINT: non-fish are Homo sapiens, Sus scrofa, and Trachemys scripta
#sdat <-

```

```

# Only keep the rows with a percent identity greater than 95
idat <- sdat[sdat$percent_id > 95, ]
# idat
# Create a new column with the genus and species called gensp
# HINT: use ?paste
idat$gensp <- paste(idat$genus, idat$species)

```

Rarefaction of samples

In order to assess the sampling depth of the eDNA sequencing run, we will first conduct a rarefaction analysis. First, we will want to sum all of the unique combinations of barcodes and gensp. Then we will create a dataframe where each row represents 1 “read” in the dataset. We will then randomly sample the dataframe and calculate the number of unique species for each sample. Finally, we will plot the rarefaction curve by barcode.

```

# Consolidate unique gensp + barcode combinations
# HINT: you will be working with the idat dataframe
#       use ?group_by, ?summarize ?arrange, ?mutate
rare <- idat %>%
  group_by(barcode, gensp) %>%
  summarize(reads = sum(reads)) %>%
  arrange(barcode, desc(reads))

```

‘summarise()’ has grouped output by ‘barcode’. You can override using the ## ‘.groups’ argument.

```

# Create a dataframe where each row corresponds to a read
# HINT: what programming construct can you use to do this?
#       how can we do this on a per-barcode basis?
#       what steps do we need to complete to do this?
#       use ?rep to get started

rarea <- data.frame()
for (i in 1:length(rare$barcode)) {
  newrows <- data.frame(
    barcode <- c(rep(rare$barcode[i], rare$reads[i])),
    gensp <- c(rep(rare$gensp[i], rare$reads[i]))
  )
  rarea <- rbind(rarea, newrows)
}

# rarea
# Randomly sample dataframe and calculate the number of unique species
# for each sample
# HINT: a nested for loop will be useful here...
rarefaction <- data.frame()
steps <- 10
minreads <- 0
for (barcode in unique(rarea$barcode)){
  barcode_reads <- rarea[rarea$barcode == barcode, ]
  numrows_barcode_reads <- nrow(barcode_reads)
  for (i in seq(1, numrows_barcode_reads, steps)){

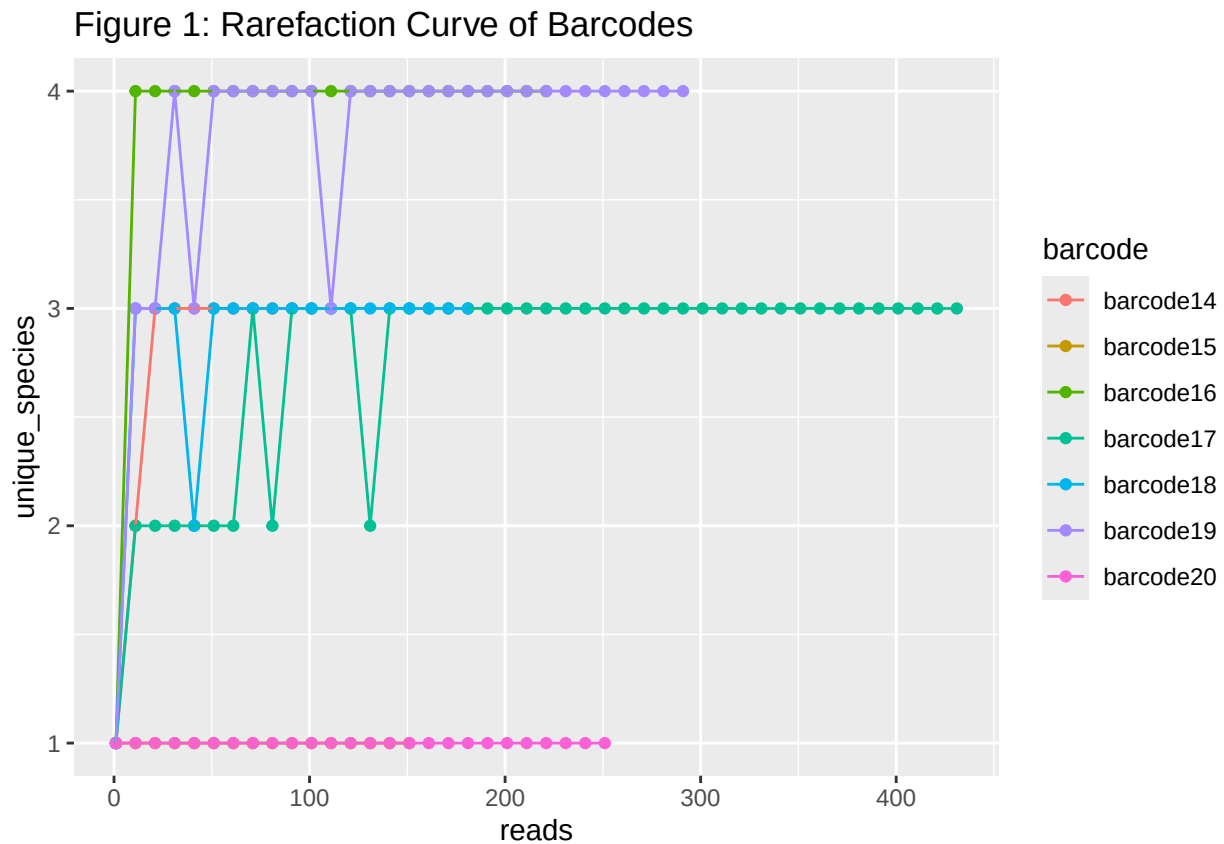
```

```

sample <- rare_sample(barcode_reads, i)
rarefaction <- rbind(rarefaction, data.frame(
# Rename the columns to the following: "unique_species", "reads", "barcode"
  barcode = barcode,
  reads = i,
  unique_species = sample)
)
}
}
rareplot <- ggplot(rarefaction, aes(x = reads, y= unique_species, color = barcode)) +
  geom_point() +
  geom_line()+
  ggtitle("Figure 1: Rarefaction Curve of Barcodes")
# Plot the rarefaction curve by barcode

# Display the rarefaction curve
rareplot

```



Comparing relative abundance and unique species identified between site replicates

To compare the consistency of the replicates of the same site, we grouped the data by barcode and species and calculated the sum of reads for each group. Then, we calculated the relative abundance of each species within each barcode group and stored the results in a new data frame called `relab`. Relative abundance

in this case is the specific species divided by the total number of species sampled. Finally, we generated a stacked bar plot using `ggplot2` library, where the x-axis represents the barcode, the y-axis represents the relative abundance of each species, and the fill color represents the species/genus.

```
# Calculate relative abundance
# Combine rows with the same genus and species
# HINT: you will want to group by barcode and gensp
#       use ?group_by, ?summarise
idat <- idat %>%
  group_by(barcode, gensp) %>%
  summarise(reads = sum(reads)) %>%
  arrange(barcode, desc(reads))
```

```
## 'summarise()' has grouped output by 'barcode'. You can override using the
## '.groups' argument.
```

```
idat
```

```
## # A tibble: 19 x 3
## # Groups:   barcode [7]
##   barcode gensp reads
##   <chr>   <chr> <int>
## 1 barcode14 Poecilia sphenops 77
## 2 barcode14 Amatitlania nigrofasciata 30
## 3 barcode14 Micropterus dolomieu 7
## 4 barcode15 Poecilia sphenops 159
## 5 barcode16 Amatitlania nigrofasciata 73
## 6 barcode16 Ancistrus temminckii 60
## 7 barcode16 Poecilia sphenops 56
## 8 barcode16 Micropterus dolomieu 33
## 9 barcode17 Ancistrus temminckii 319
## 10 barcode17 Amatitlania nigrofasciata 109
## 11 barcode17 Awaous stamineus 7
## 12 barcode18 Ancistrus temminckii 133
## 13 barcode18 Amatitlania nigrofasciata 49
## 14 barcode18 Micropterus dolomieu 7
## 15 barcode19 Ancistrus temminckii 188
## 16 barcode19 Amatitlania nigrofasciata 90
## 17 barcode19 Poecilia sphenops 13
## 18 barcode19 Micropterus dolomieu 6
## 19 barcode20 Micropterus dolomieu 255
```

```
# Calculate the total number of reads
# HINT: we will want to do this on a per-barcode basis
bc <- c()
for(barcode in unique(idat$barcode)){
  bc <- c(bc, sum(idat[idat$barcode == barcode,]$reads))
}
names(bc) <- unique(idat$barcode)
bc
```

```
## barcode14 barcode15 barcode16 barcode17 barcode18 barcode19 barcode20
##      114      159      222      435      189      297      255
```

```

# Calculate the relative abundance
# HINT: what programming construct can you use to do this?
relab <- data.frame()
for(i in 1:nrow(idat)){
  relab[i,"barcode"] <- idat$barcode[i]
  relab[i,"gensp"] <- idat$gensp[i]
  relab[i,"relative_abundance"] <- idat$reads[i] / bc[idat$barcode[i]]
}

relab

```

```

##      barcode      gensp relative_abundance
## 1 barcode14 Poecilia sphenops      0.67543860
## 2 barcode14 Amatitlania nigrofasciata      0.26315789
## 3 barcode14 Micropterus dolomieu      0.06140351
## 4 barcode15 Poecilia sphenops      1.00000000
## 5 barcode16 Amatitlania nigrofasciata      0.32882883
## 6 barcode16 Ancistrus temminckii      0.27027027
## 7 barcode16 Poecilia sphenops      0.25225225
## 8 barcode16 Micropterus dolomieu      0.14864865
## 9 barcode17 Ancistrus temminckii      0.73333333
## 10 barcode17 Amatitlania nigrofasciata      0.25057471
## 11 barcode17 Awaous stamineus      0.01609195
## 12 barcode18 Ancistrus temminckii      0.70370370
## 13 barcode18 Amatitlania nigrofasciata      0.25925926
## 14 barcode18 Micropterus dolomieu      0.03703704
## 15 barcode19 Ancistrus temminckii      0.63299663
## 16 barcode19 Amatitlania nigrofasciata      0.30303030
## 17 barcode19 Poecilia sphenops      0.04377104
## 18 barcode19 Micropterus dolomieu      0.02020202
## 19 barcode20 Micropterus dolomieu      1.00000000

```

```

# Plot the relative abundance
relab_plot <- ggplot(data = relab, aes(x = barcode, y = relative_abundance, fill = gensp)) +
  geom_bar(stat = "identity", position = "stack") +
  xlab("barcode")+
  ylab("relative abundance")+
  ggtitle("Figure 2:Relative Abundance of Stream Data")

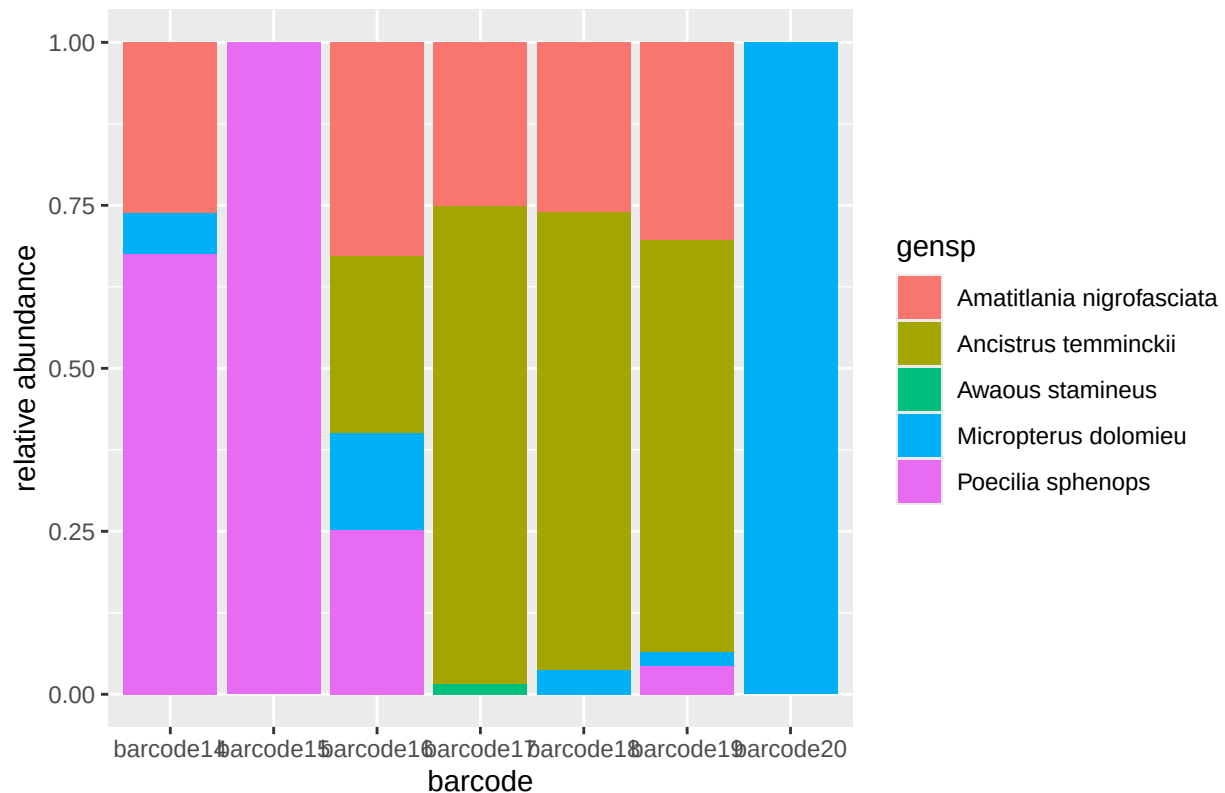
```

```

# Display the relative abundance plot
relab_plot

```

Figure 2: Relative Abundance of Stream Data



Comparing relative abundance and unique species of pooled site replicates to the survey data

To compare the sequencing data to the survey data, we first created a new dataframe for the counts from survey data and the reads from sequencing data. However, instead of keeping the sequencing data as separate replicates we combined them into a single group. We then generated a stacked bar plot using `ggplot2` library, where the x-axis represents the sampling method, the y-axis represents the relative abundance of each species, and the fill color represents the species.

```
# Make new dataframe for compaision to survey data
# HINT: you will want to group by gensp
#       use ?group_by, ?summarise
sc <- idat %>%
  group_by(gensp) %>%
  summarise(reads = sum(reads)) %>%
  arrange(barcode, desc(reads))

# Add a column for the method
sc$method <- "sequencing"

# Prepare the survey data
# HINT: we will want to replace the spaces in the species names with underscores
#       and create a new column for the gensp
#       use ?str_replace_all
```

```
sc$gensp <- str_replace_all(sc$gensp, " ", "_")
sc$gensp
```

```
## [1] "Ancistrus_temminckii"      "Amatitlania_nigrofasciata"
## [3] "Micropterus_dolomieu"     "Poecilia_sphenops"
## [5] "Awaous_stamineus"
```

```
# Summarize the survey data
# HINT: we will want to group by gensp
#       use ?group_by, ?summarise
sur$gensp <- str_replace_all(sur$Species, " ", "_")
sur$gensp
```

```
## [1] "Amatitlania_nigrofasciata" "Xiphophorus_hellerii"
## [3] "Poecilia_sphenops"        "Poecilia_reticulata"
## [5] "Ancistrus_temminckii"
```

```
# Add a column for the method
sur$method <- "surveying"
surdat <- subset(sur, select = c(gensp, Count, method))
surdat <- surdat %>%
  rename( reads = Count)
surdat
```

```
##               gensp reads    method
## 1 Amatitlania_nigrofasciata  44 surveying
## 2      Xiphophorus_hellerii    2 surveying
## 3      Poecilia_sphenops     4 surveying
## 4      Poecilia_reticulata  194 surveying
## 5      Ancistrus_temminckii   13 surveying
```

```
# Combine the two dataframes
relab_compare <- rbind(surdat,sc)
arrange(relab_compare, desc(reads))
```

```
##               gensp reads    method
## 1      Ancistrus_temminckii  700 sequencing
## 2 Amatitlania_nigrofasciata  351 sequencing
## 3      Micropterus_dolomieu  308 sequencing
## 4      Poecilia_sphenops    305 sequencing
## 5      Poecilia_reticulata  194  surveying
## 6 Amatitlania_nigrofasciata   44  surveying
## 7      Ancistrus_temminckii   13  surveying
## 8      Awaous_stamineus       7 sequencing
## 9      Poecilia_sphenops      4  surveying
## 10     Xiphophorus_hellerii    2  surveying
```

```
# Calculate the relative abundance for the survey data and sequencing data
# HINT: what programming construct can you use to do this?
```

```
total_reads <- c()
```

```

for(i in unique(relab_compare$method)){
total_reads <- c(total_reads, sum(relab_compare[relab_compare$method == i, ]$reads))
}
names(total_reads) <- unique(relab_compare$method)

relab_abundance <- data.frame()

for(i in 1:nrow(relab_compare)){
relab_abundance[i, "gensp"] <- relab_compare$gensp[i]
relab_abundance[i, "method"] <- relab_compare$method[i]
if(relab_compare[i, "method"] == "surveying"){
relab_abundance[i, "relative_abundance"] <- relab_compare$reads[i]/total_reads[1]
}else{
  relab_abundance[i, "relative_abundance"] <- relab_compare$reads[i]/total_reads[2]
}
}
relab_abundance

```

```

##              gensp      method relative_abundance
## 1  Amatitlania_nigrofasciata surveying      0.171206226
## 2      Xiphophorus_hellerii surveying      0.007782101
## 3      Poecilia_sphenops surveying      0.015564202
## 4      Poecilia_reticulata surveying      0.754863813
## 5      Ancistrus_temminckii surveying      0.050583658
## 6      Ancistrus_temminckii sequencing      0.418910832
## 7  Amatitlania_nigrofasciata sequencing      0.210053860
## 8      Micropterus_dolomieu sequencing      0.184320766
## 9      Poecilia_sphenops sequencing      0.182525434
## 10      Awaous_stamineus sequencing      0.004189108

```

```

# # Plot the relative abundance
# relab_plot <- ggplot(data = relab, aes(x = barcode, y = relative_abundance, fill = gensp)) +
#   geom_bar(stat = "identity", position = "stack") +
#   xlab("barcode")+
#   ylab("relative abundance")

```

```

# Plot the relative abundance
relab_compare_plot <- ggplot(data = relab_abundance, aes(x = method, y= relative_abundance, fill = gensp)) +
  geom_bar(stat = "identity", position = "stack")+
  xlab("method")+
  ylab("relative abundance")+
  ggtitle("Figure 3: Relative Abundance of Species in each method of Sampling")

```

```

# Display the relative abundance comparison plot
relab_compare_plot

```

Determining relationships between sequencing reads and survey counts

To determine the relationship between the sequencing reads and survey counts, we first created a new dataframe for the counts from survey data and the reads from sequencing data. We then created a linear

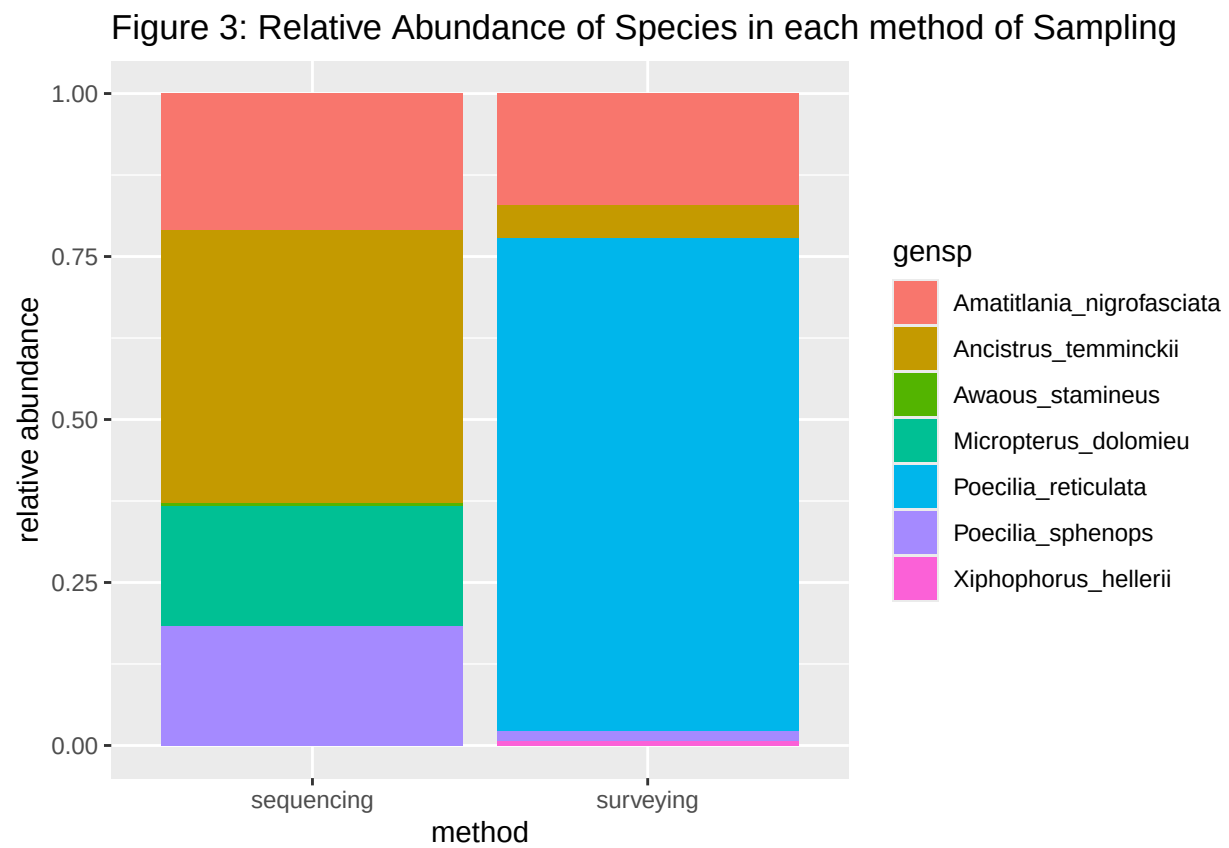


Figure 1: Relative abundance of species between sequencing reads and survey counts.

model to compare the counts from survey data and the reads from sequencing data. Finally, we generated a scatter plot using `ggplot2` library, where the x-axis represents the survey counts, the y-axis represents the sequencing reads, and the color represents the species.

```
# Make 2 new dataframes for the counts from survey data and the reads from sequencing data
# Assign the names reads and counts to the numeric columns
survey_counts <- data.frame(gensp = surdat$gensp, count = surdat$reads, method = "surveying")

# Merge the two dataframes
# HINT: use ?merge
comb3 <- merge(sc,survey_counts, by = c("gensp", "method"), all = TRUE)

# Replace NA values with 0
comb3$reads[is.na(comb3$reads)] = 0
comb3$count[is.na(comb3$count)] = 0
# Create linear model to compare the counts from survey data and the reads from
# sequencing data
# All of the code below is a function that will generate the equation for the linear model
# HINT: you will need to complete the function
lm_eqn <- function(df) {
  m <- lm(reads ~ count , df);
  eq <- substitute(italic(y) == a + b %.% italic(x)*",~~italic(r)^2~"=~r2,
    list(a = format(unname(coef(m)[1]), digits = 2),
          b = format(unname(coef(m)[2]), digits = 2),
          r2 = format(summary(m)$r.squared, digits = 3)))
  as.character(as.expression(eq));
}

# Plot the counts from survey data and the reads from sequencing data
seqvsur_lm_plot <- ggplot(data = comb3, aes(x = count, y = reads))+
  geom_text(x = 35, y = 4000, label = lm_eqn(comb3), parse = TRUE)+
  geom_point(aes(color = gensp))+
  coord_fixed(ratio = 0.2)+
  geom_smooth(method = "lm", se = FALSE)+
  ggtitle("Figure 4: Relationship between eDNA reads and survey count")

# lm_model <- lm(reads~count, data = comb3)
# summary(lm_model)

# Display the linear model plot
seqvsur_lm_plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Conclusions

The primary findings of the analysis are as follows: The database generated for this analysis must be robust enough to identify all species of interest; if the database is not robust it may result in misclassifications. That is why we filtered the data to include species with a percent identity of higher than 95%. In addition, we used a rarefaction curve so that we know how much times the data needs to be sampled to have an accurate representation of the stream population(Figure 1). In the relative abundance graph(Figure 2), it tells us the species composition in the Manoa Stream where *Micropeterus dolomieu* and *Poecilia sphenops*

Figure 4: Relationship between eDNA reads and survey count

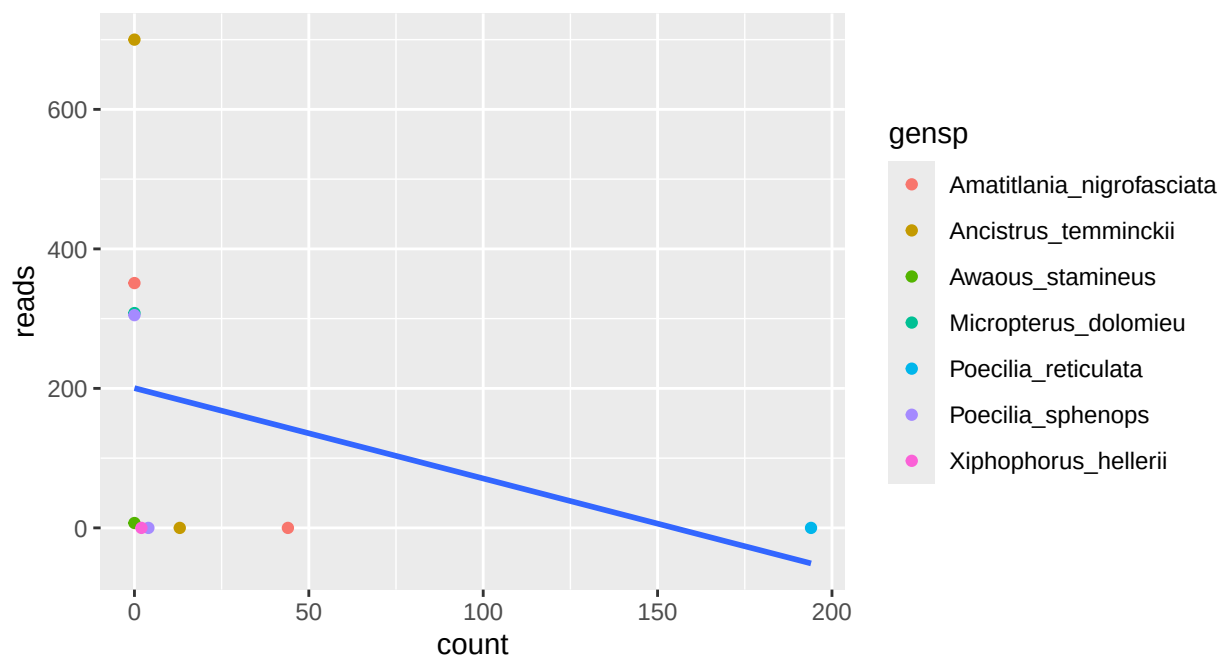


Figure 2: Relationship between sequencing reads and survey counts.

have highest relative abundance. This means that the stream population has a high composition of both invasive species and constantly compete against other native and invasive species in the same population. The relative abundance graph in (Figure 3) determines the difference in sampling between a manual survey and using eDNA. We can see a high relative abundance for *Poecilia Reticulata* when manually surveying the Manoa stream as opposed to sampling via eDNA. This difference could've happened because we are randomly sampling a large body of water where it could contain more eDNA of a certain species. Similarly, this could happen when manually surveying the population since the biologist is only surveying a fraction of the stream. The graph from (Figure 4) shows the counts for survey vs the reads from the eDNA sampling. An r^2 value of -0.05 means that the 5% of the variability in the counts is explained by the reads. This proves that they are not correlated to each other. We can conclude that eDNA cannot be used as a replacement for manual surveys. However, it can be used as a supplement to manual surveys and as a tool to determine if a species is present in a given area and allows for detection of species that were not captured in the manual survey.

Sources

<https://www.ctahr.hawaii.edu/brulandg/whatsnew/HawaiiWetlandMonitor200904.pdf>

[https://dlnr.hawaii.gov/ais/other-ais/guppies/#:~:text=Guppies%20\(Poecilia%20reticulata\)%20are%20small%20freshwater](https://dlnr.hawaii.gov/ais/other-ais/guppies/#:~:text=Guppies%20(Poecilia%20reticulata)%20are%20small%20freshwater)